Swiss dynamic tariffs

What do we need?

- API definitions
- What do we do, if an API is not reachable?
- What happens if there is no new data?

API definitions

- REST APIs
- Time resolution is always 15 minutes and timezone is UTC
- **API 1: What dynamic tariffs are available per provider today?**

```
All data at once without parameters

return:
  tariff_id
  tariff_name
  provider_name
  Where?
    ZIP range (list of ZIP ids)
  (What? -> unused, because we expect all components energy, grid, residual)
  When? -> When is the daily update in UTC?
    daily_update_time_utc
  Available since? -> What is the first time, we can get values for this tariff
    datetime_available_from_utc
  Valid until [optional] -> Tariff is not supported anymore, None if ongoing
```

- **API 2: timeseries data in 15 min intervals and timezone UTC.**

```
API parameters:
  tariff_id
  start_time: arbitrary date in the past in UTC

optional API parameters:
  end_time: in UTC

return: timeseries of netto prices per kWh in the following format
  energy_price_utc:
    list:
      {start_time: netto_price}
      Example:
      {start_timestamp (UTC): netto_price (CHF/kWh)},
      {start_timestamp (UTC) + 1x 15min: netto_price2 (CHF/kWh)},
      {start_timestamp (UTC) + 2x 15min: netto_price2 (CHF/kWh)},
      ...
      {end_timestampe (UTC) - 15min), netto_price_end (CHF/kWh)}
  grid_price_utc:
    list:
      {start_time: netto_price}
      Example:
      {start_timestamp (UTC): netto_price (CHF/kWh)},
      {start_timestamp (UTC) + 1x 15min: netto_price2 (CHF/kWh)},
      {start_timestamp (UTC) + 2x 15min: netto_price2 (CHF/kWh)},
      ...
      {end_timestampe (UTC) - 15min), netto_price_end (CHF/kWh)}
  residual_price_utc: # "sonstige Kosten, SDL, Stromreserve, Netz", Messtarif, Zuschlaege, Abgaben, ..."
    list:
      {start_time: netto_price}
      Example:
      {start_timestamp (UTC): netto_price (CHF/kWh)},
      {start_timestamp (UTC) + 1x 15min: netto_price2 (CHF/kWh)},
      {start_timestamp (UTC) + 2x 15min: netto_price2 (CHF/kWh)},
      ...
      {end_timestampe (UTC) - 15min), netto_price_end (CHF/kWh)}

```

Further Assumption and clarifications

- Error handling: No data is returned at the expected time?
  - No new data or not responding: After 1 hour we send a Mail

- Please take extra care of daylight saving time!
- We are only interested in costs per energy unit and not in monthly/yearly costs.

Open Question

- Are all coupled tariffs used mandiatorily or is the coupling optional?
- All prices do not include residual tariff components (that are neither grid costs nor energy costs), e.g. Stromsteuer, ...?

Sources

- Dynamische Tarife – smartgridready

Deprecated

Alternative for return (@Hadi Samadi decided against this option. fyi @Markus Placho)

```
return:
  energy_price_utc:
    list:
      {start_time: {'energy_price': netto_price_energy, 'grid_price': ne
```